Reconstructing Suspected Drug Trafficking From Chrome Web History

Case Study 2 Individual browser history investigation

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1. Case reference and student identification

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Registration number	2025/FWSD/11202
Course code	SBT-DF204 Computer Forensics Case Studies
Case title	Reconstructing Suspected Drug Trafficking From Chrome Web History
Evidence examined	Google Chrome History SQLite database, supplied as Chrome_History_Cryptocurrency_Lab.db (profile of Windows user FSCS_User)
Examination date	11 October 2026
Submission file name	SBTDF204_CaseStudy2_2025FWSD11202.docx

2. Executive conclusion

The History database (supplied as Chrome_History_Cryptocurrency_Lab.db) records 74 visits to 57 URLs, all between **13:54:18 and 15:02:56 UTC on 19 April 2022**, plus one completed download. The records form a coherent, time-ordered sequence that is **consistent with** the case hypothesis: a Craigslist account log-in and a multi-step posting workflow producing a listing titled "cheaper than Rx supplements" (13:54–13:58 UTC); access to a Gmail account and an email thread with the same subject followed by "order" (14:04 UTC); a later thread whose subject contains a 64-character Bitcoin-style transaction ID (14:56 UTC); an Imgur page reached from that email and a download named proof_of_payment.png (14:56–14:57 UTC); lookup of the same transaction ID in two blockchain explorers and a Kraken exchange session (14:58–15:02 UTC).

The history proves that this Chrome profile requested these URLs in this order. It does **not** prove the content of any page or email, that a drug was offered or sold, that the payment was real or received, or who was at the keyboard. The only drug-related text in the database is the listing title "Rx supplements" in the "health and beauty" category; no record shows a controlled substance. **Confidence:** moderate-to-high that the profile performed the posting-to-payment-verification sequence; low-to-moderate that it amounts to drug trafficking; low that it identifies a named individual (Section 7).

3. Evidence acquisition and integrity

The case file is a Google Chrome history database. The assessment brief calls it History and states that it is "added in study material section"; it was supplied through the course study material and saved to the Downloads folder of the examiner's Kali Linux workstation on 11 October 2026 under the file name Chrome_History_Cryptocurrency_Lab.db.

Original file name	Chrome_History_Cryptocurrency_Lab.db (the brief's History file)
Source	Course study material for SBT-DF204 Case Study 2 (downloaded to ~/Downloads)
File size	196,608 bytes
SHA-256 (original)	b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19
SHA-256 (working copy)	b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19 (identical)
Read-only original	~/SBTDF204_CaseStudy2/01_original_evidence/Chrome_History_Cryptocurrency_Lab.db (mode 444)
Working copy	~/SBTDF204_CaseStudy2/02_working_copy/Chrome_History_Cryptocurrency_Lab.db
Tools	Kali Linux; cp, chmod, stat, sha256sum; sqlite3 command-line shell

Working-copy method. An assignment folder was created first, with separate folders for the original evidence, working copy, SQL, screenshots, notes and submission (Figure 1). The file was copied with cp -p into 01_original_evidence/ and made read-only with chmod 444. Its size was read with stat and its SHA-256 with sha256sum. A second copy was placed in 02_working_copy/ and hashed; the two hashes are identical (Figure 2). Every query was run only against the working copy with sqlite3 -readonly, so the evidence could not be modified. Each action was recorded with a UTC timestamp in a case log (05_notes/case_log.txt).

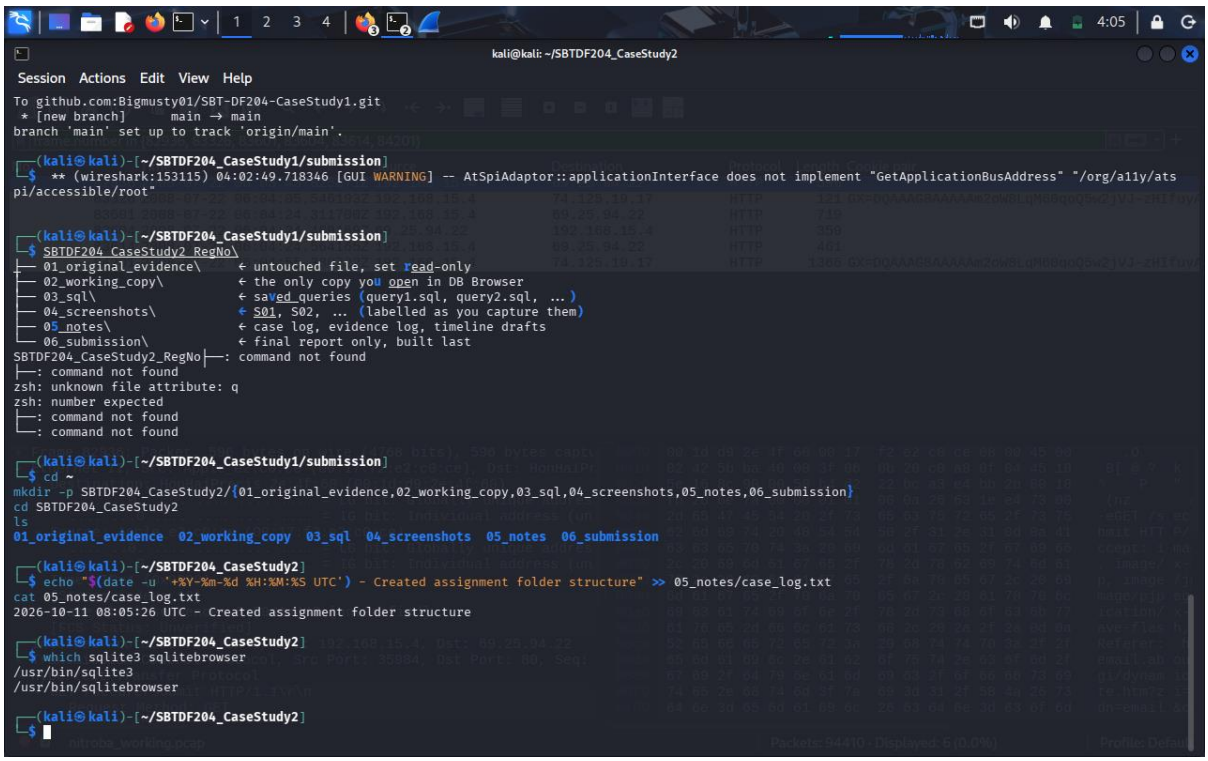


Figure 1 *Assignment folder structure created, first case-log entry, and* sqlite3 *and* sqlitebrowser *confirmed installed.*

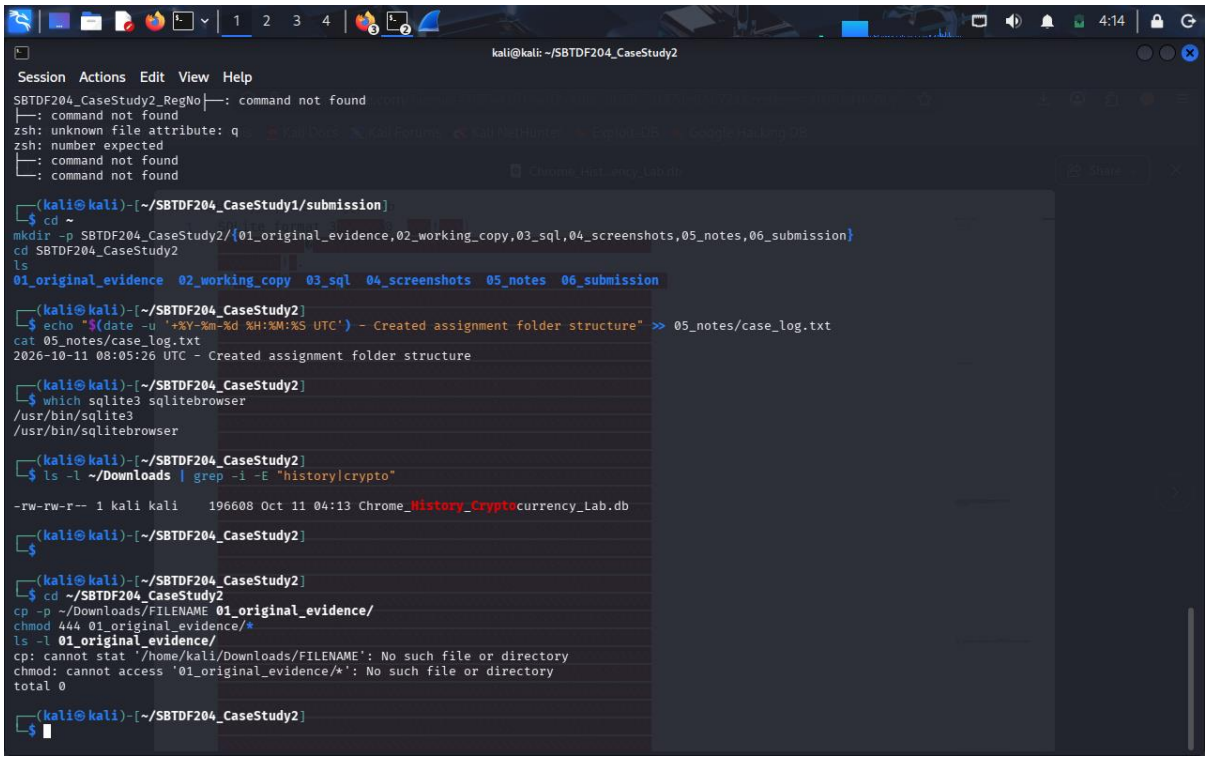


Figure 1a *Source file located in* ~/Downloads *(196,608 bytes), followed by a first copy attempt that failed because a placeholder file name (*FILENAME*) was typed; no file was created or altered (*total 0*). The correct copy is shown in Figure 2.*

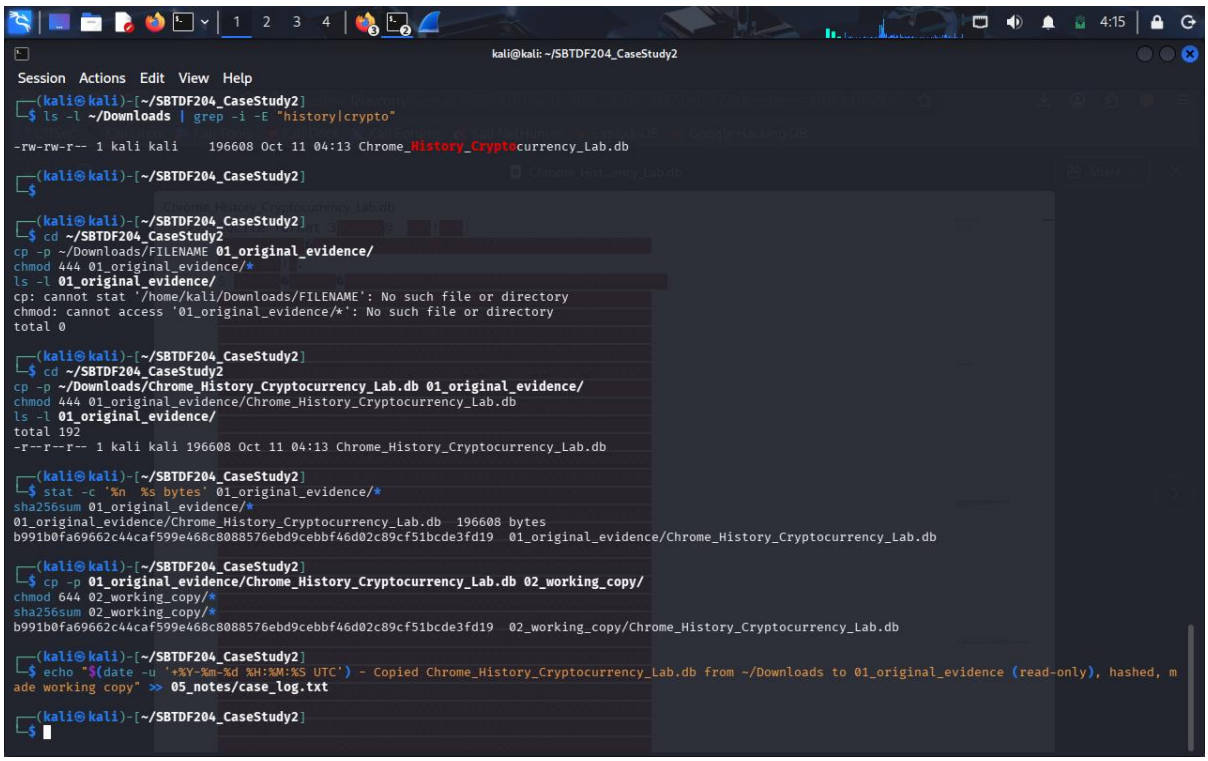


Figure 2 *Original copied to* 01_original_evidence *(read-only, 196,608 bytes); SHA-256 of the original and the working copy are identical; case-log entry written (E01).*

Precautions. No live site, account, e-mail, exchange, wallet or transaction ID referenced in the history was visited or queried. The Chrome meta table reports history schema version 53 (last compatible version 16).

4. Database method

4.1 Tables reviewed

The schema was listed first (sqlite_master). Tables holding data were: urls (57 rows), visits (74), downloads (1), downloads_url_chains (1), keyword_search_terms (4), content_annotations (47) and context_annotations (45). visit_source, segments, segment_usage, clusters, clusters_and_visits, downloads_slices, downloads_reroute_info and typed_url_sync_metadata were empty.

Table / field	Role in this case
urls	One row per distinct URL: id, url, title, visit_count, typed_count, last_visit_time, hidden. Titles supply the page names relied on (e.g. "posting confirmation", "Inbox (12) - …", "Transaction: …"). typed_count is 0 for every URL.
visits	One row per page load: id, url (→ urls.id), visit_time (WebKit µs), from_visit (the visit that led here), transition (type + qualifier bits), visit_duration (µs), opener_visit (visit that opened this one in a new tab/window).
downloads	File-save record: target path, start/end time, bytes received/total, state, referrer, tab_url, etag, last_modified, MIME type.
downloads_url_chains	The URL(s) actually fetched for a download, in order (chain_index); here the direct image URL on i.imgur.com.
keyword_search_terms	Search terms entered through the omnibox/search box (four terms: craigslist, gmail, blockchain explorer, kraken), tied to urls.id.

4.2 Timestamp conversion

Chrome stores visit_time, start_time and end_time as microseconds since 1 January 1601 00:00:00 UTC. They were converted with datetime(t/1000000 + strftime('%s','1601-01-01'), 'unixepoch'), which yields UTC; all times in this report are UTC. Example: visit 1, visit_time 13294850058231547 → 2022-04-19 13:54:18 UTC. Durations in visit_duration are microseconds, shown here in seconds. If the device clock was set to US Eastern time (EDT, UTC−4 on that date an assumption suggested by the Baltimore Craigslist site and the en-US search parameter rlz=…enUS989US989, not proven), local times would read four hours earlier (e.g. 09:54:18). No local-time conversion is used in the tables.

4.3 Joins and transition interpretation

Visits were joined to URLs on visits.url = urls.id and ordered by visit_time (Appendix A, query4_timeline.sql). The base transition is transition & 0xFF; qualifier bits are read from the high bits of the unsigned 32-bit value (SQLite returns some as negative integers, e.g. −2147483648 = 0x80000000). The values present in this database are:

Stored value	Hex	Base type	Qualifiers	Count	Visit IDs
805306373	0x30000005	5 = GENERATED	start of a navigation chain; end of a chain	1	1
805306368	0x30000000	0 = LINK	start of a navigation chain; end of a chain	21	2, 20, 21, 23, 34, 35, 36, 47, …
268435456	0x10000000	0 = LINK	start of a navigation chain	12	3, 6, 9, 11, 24, 31, 37, 43, …
-2147483648	0x80000000	0 = LINK	HTTP server redirect	13	4, 12, 25, 26, 27, 28, 29, 32, …
-1610612736	0xa0000000	0 = LINK	end of a chain; HTTP server redirect	9	5, 7, 10, 13, 30, 33, 50, 56, …
805306375	0x30000007	7 = FORM_SUBMIT	start of a navigation chain; end of a chain	7	8, 14, 15, 16, 17, 18, 19
838860805	0x32000005	5 = GENERATED	initiated from the address bar; start of a navigation chain; end of a chain	3	22, 51, 59
1610612736	0x60000000	0 = LINK	end of a chain; client (script/meta) redirect	3	42, 46, 74
1073741824	0x40000000	0 = LINK	client (script/meta) redirect	5	44, 45, 68, 72, 73

Reading rules applied: (a) base 5 (GENERATED) with FROM_ADDRESS_BAR = a search entered in the address bar (visits 22, 51, 59; visit 1 lacks the address-bar bit, so it came from another omnibox/search-box path); (b) base 7 (FORM_SUBMIT) = a form was submitted to reach the page (visits 8, 14–19); (c) from_visit was followed to rebuild chains, and opener_visit to identify pages opened in a new tab or window; (d) redirect qualifiers were treated as automatic hops, not user actions. There are no base-1 (TYPED) transitions and every typed_count is 0, so no URL was recorded as typed in full; the user-initiated entry points are the four omnibox searches.

5. Findings

Q1 Evidence file and preservation

History, 196,608 bytes, SHA-256 b991b0fa…cde3fd19, supplied as Chrome_History_Cryptocurrency_Lab.db and preserved as described in Section 3 (evidence log **E01**, Figures 1–2). The integrity check passed (**E02**, Figure 3).

Q2 Relevant tables and fields

See Section 4.1. The urls ↔ visits join provides what was visited, when and how; downloads + downloads_url_chains document the single file saved (**E02, E12**; Figures 3 and 10).

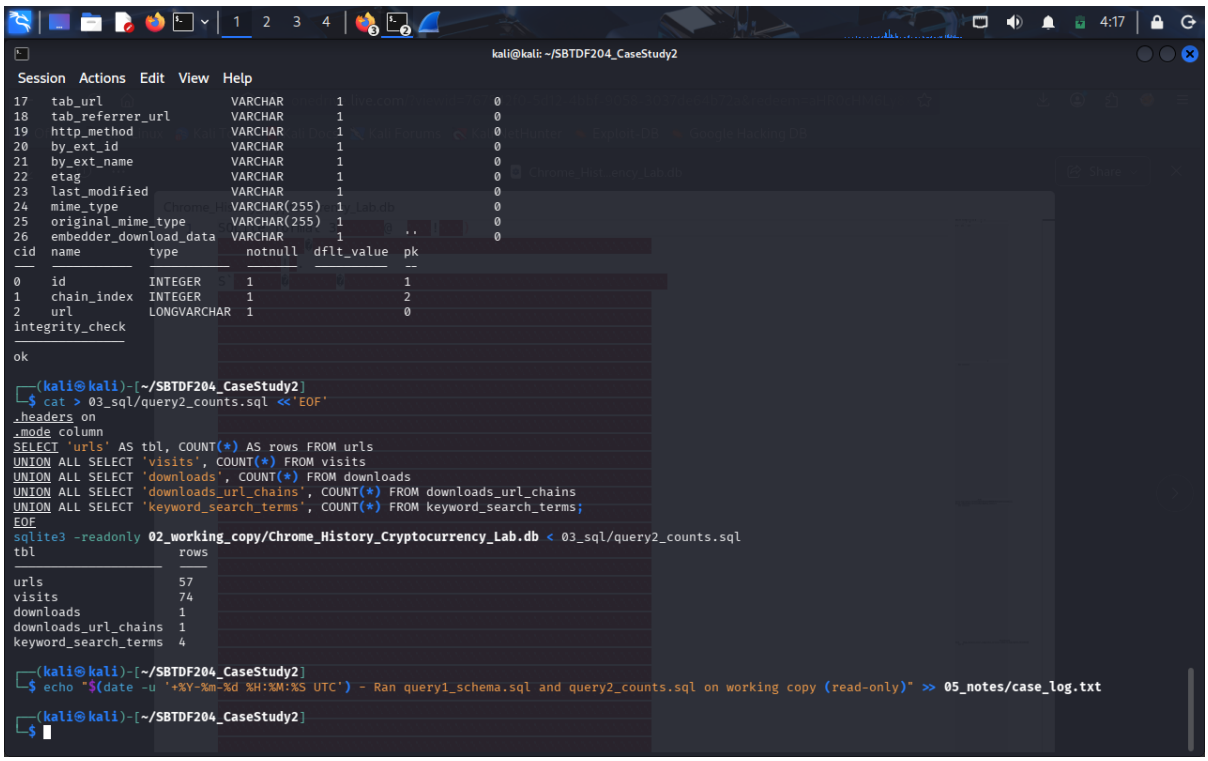


Figure 3 *End of the schema listing (*downloads_url_chains *fields),* PRAGMA integrity_check *= ok, and row counts: urls 57, visits 74, downloads 1, downloads_url_chains 1, keyword_search_terms 4 (E02).*

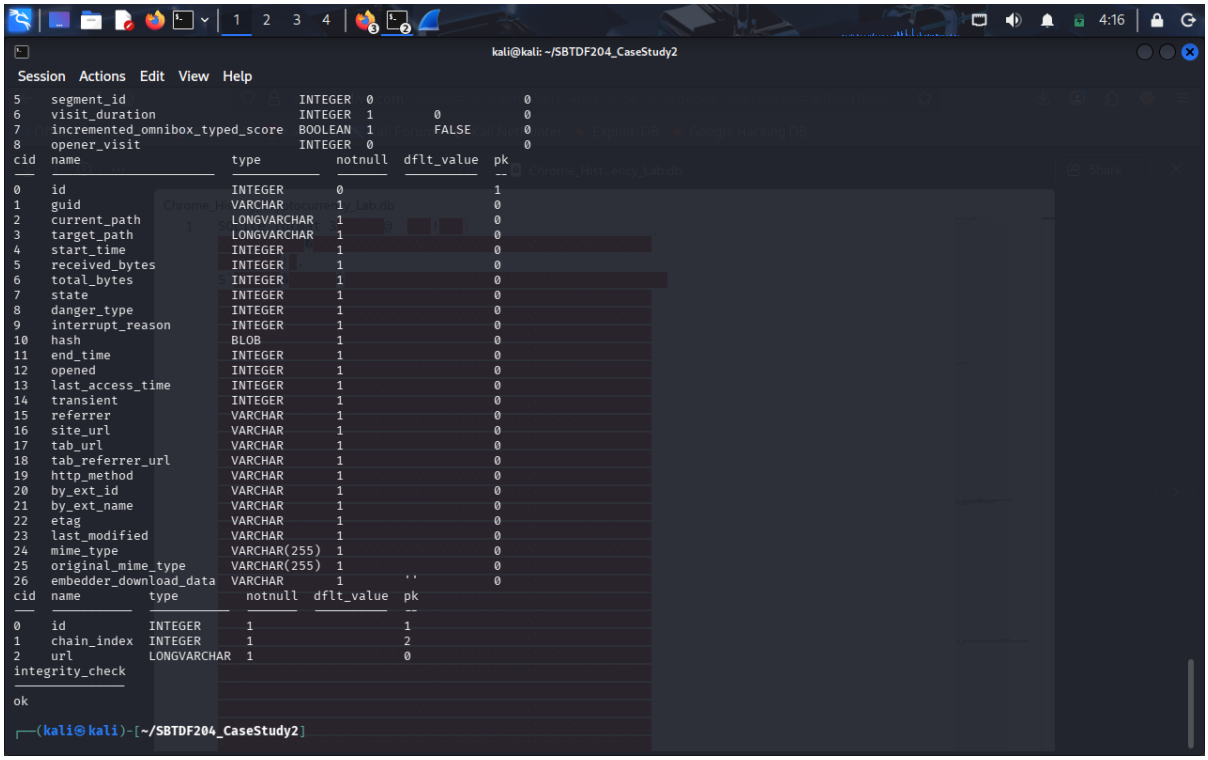


Figure 3a *Earlier part of the* query1_schema.sql *output:* PRAGMA table_info *fields of* visits *(including* from_visit, visit_duration, opener_visit*) and all 27 fields of* downloads, *then* downloads_url_chains *and the integrity check (E02).*

Q3 When the activity occurred

First visit: visit 1, 2022-04-19 13:54:18 UTC. Last visit: visit 74, 15:02:56 UTC. Total span 1 h 08 min 38 s, with one long gap (visit 47 at 14:05:59 → visit 48 at 14:56:47, about 50.8 minutes) between the "order" email thread and the

transaction-ID thread. The download ran 14:56:58 → 14:57:10 UTC (11.8 s). Method: Section 4.2 with query3_timerange.sql and query4_timeline.sql (**E03**; first visit visible in Figure 4, last visit in Figure 11).

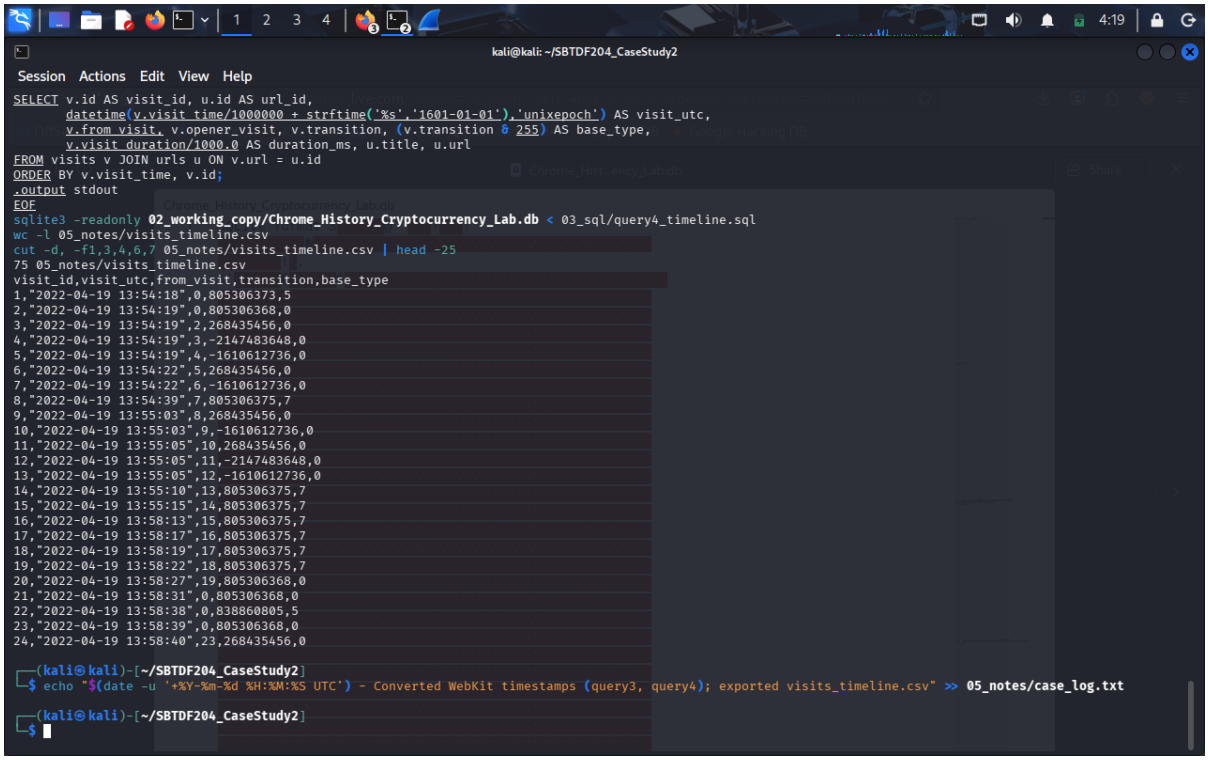


Figure 4 query4_timeline.sql*: timeline exported to* visits_timeline.csv *(75 lines = 74 visits + header). Visit 1 is at 2022-04-19 13:54:18 UTC (E03).*

Q4 How the browser reached the material URLs

- **Search entry.** Visits 1, 22, 51 and 59 are base-5 omnibox searches for "craigslist", "gmail", "blockchain explorer" and "kraken"; visits 22, 51 and 59 carry the address-bar qualifier. Each is followed by a second visit to the same results URL recorded as a link with opener = the search visit (V2, V23, V52, V60), and then by a link to the destination whose from_visit is that second visit (V3, V24, V53, V61).
- **Redirect chains.** Visits 3→5 (craigslist.org → geo → baltimore.craigslist.org), 6→7 (account home → log-in page), 9→10 (www → baltimore), 11→13 (post → draft key → ?s=type), 24→30 (Gmail entry → Gmail about page), 31→33 (account chooser), 37→42 (CheckCookie → SetOSID → SetSID → mail → #inbox), 43→46 and 72→74 (Gmail ?compose= fragments), 49→50 (Gmail link wrapper → Imgur), 55→56 (explorer search → /btc/tx/) and 67→70 (Kraken redirect → mempool.space) are redirect chains, identified by the qualifier bits and from_visit links.
- **Form submissions.** Visit 8 (Craigslist log-in), visits 14–19 (posting wizard steps: category, edit, map, image, preview, confirmation). Each has from_visit = the preceding wizard step.
- **Openers.** Visit 21 (live listing) has from_visit 0 and opener_visit 20 (the manage-posting page), so it was opened from that page rather than navigated to in place. Visits 43, 47 and 48 (Gmail) and 62–63, 65–66, 67 and 71 (Kraken) likewise carry opener_visit values pointing to earlier visits.

Evidence: **E04–E16**; full per-visit values in Appendix C.

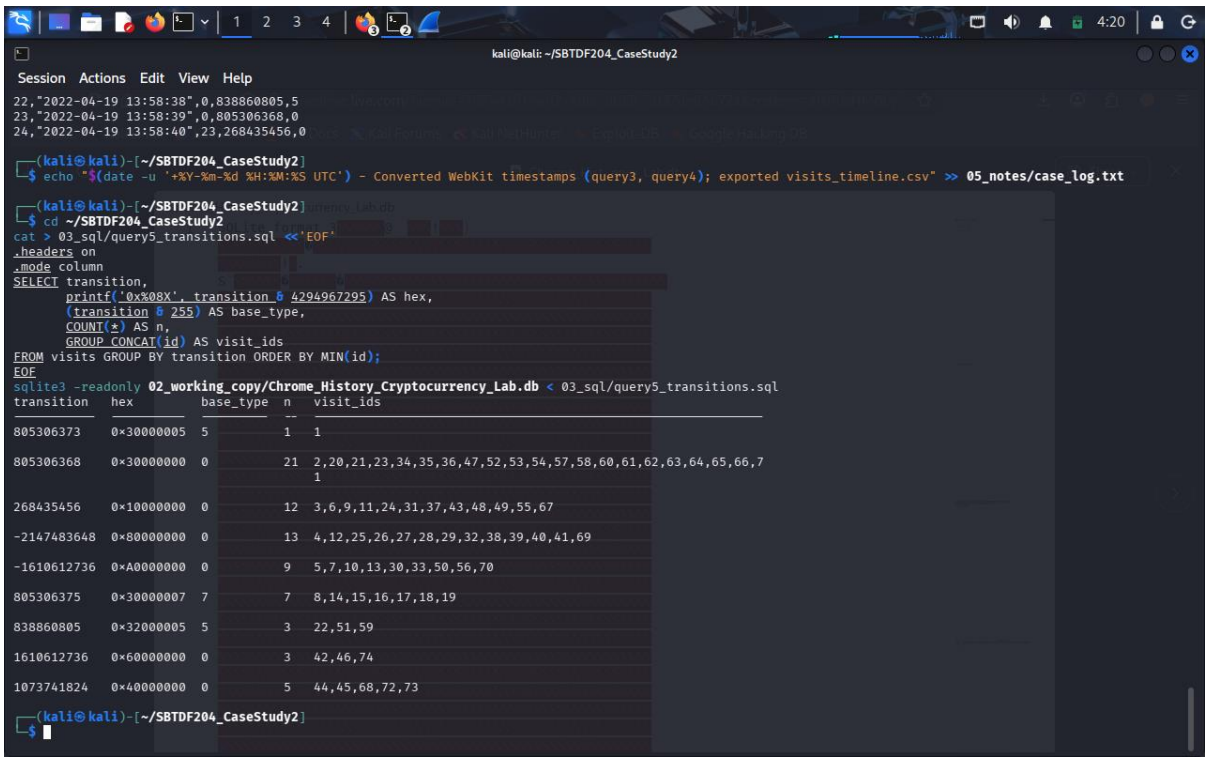


Figure 5 query5_transitions.sql*: the nine distinct transition values with hex, base type, counts and visit IDs (E04).*

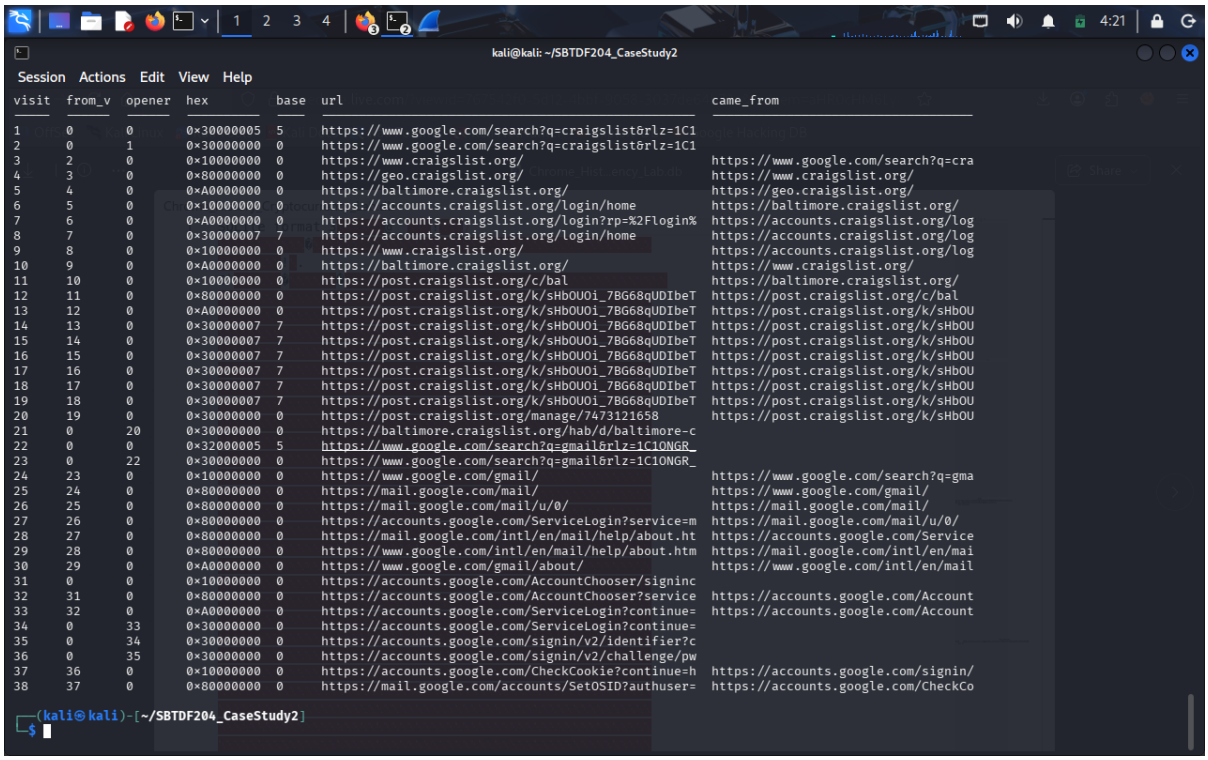


Figure 6 query6_chains.sql*: each visit with* from_visit, opener_visit*, hex transition, base type, URL and the URL it came from (visits 1–38) (E04–E06).*

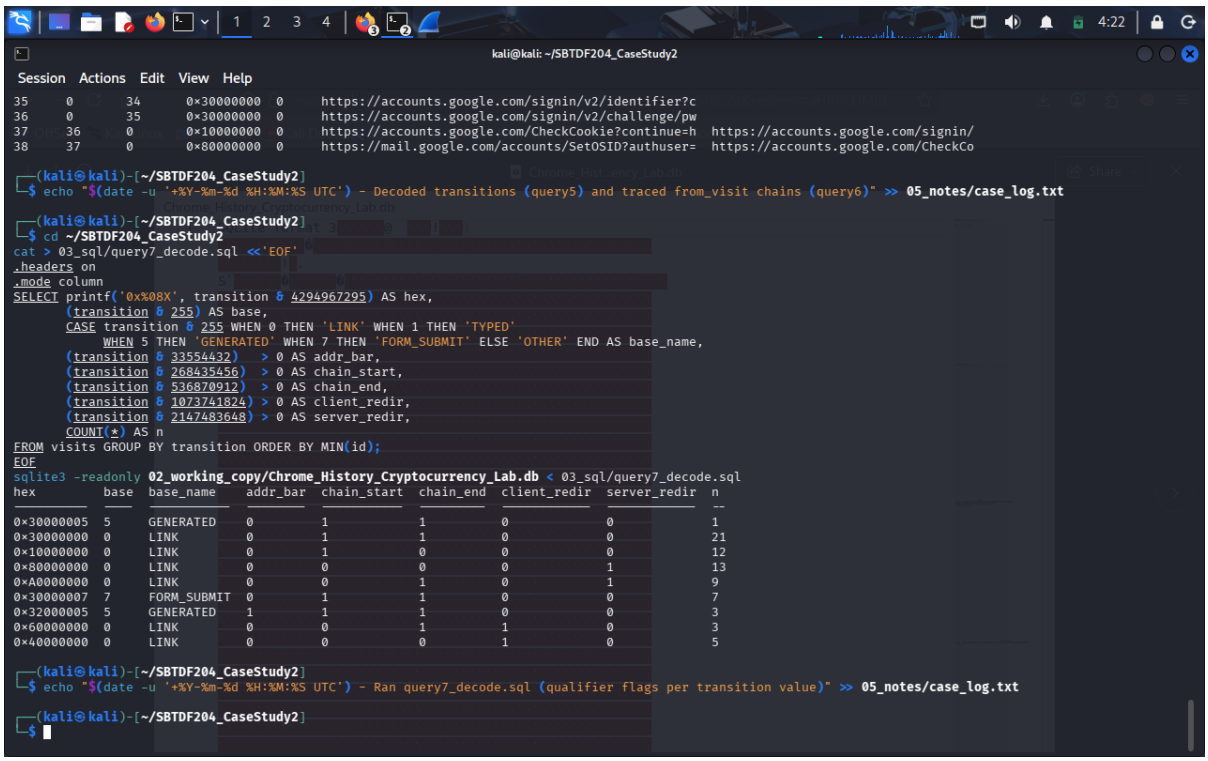


Figure 7 query7_decode.sql*: qualifier flags (address bar, chain start, chain end, client redirect, server redirect) for each transition value (E04).*

Q5 Evidence relating to the suspected posting

At 13:54:19 the profile reached craigslist.org (visits 3–5) and at 13:54:22 was sent to an account log-in page (visit 7, 17.5 s on page). Visit 8 at 13:54:39 is a base-7 form submission to accounts.craigslist.org/login/home titled "craigslist account", from_visit 7 (23.4 s on the log-in page): consistent with a log-in, though credentials and outcome are not recorded (**E05**). At 13:55:05 the posting flow began at post.craigslist.org/c/bal ("choose type"), redirecting to a draft URL carrying a session key (visits 11–13). Form-submit visits then step through "choose category" (14, 13:55:10), "create posting" (15, 13:55:15 **177.7 s** on page, matching the 13:58:13 start of the next step), "add map" (16), "create posting"/image step (17, 13:58:17), preview (18, 13:58:19) and "posting confirmation" (19, 13:58:22) the same URL (url_id 8) first seen as a redirect stop at 13:55:05, so the title alone at visit 12 does not show publication. Visit 20 at 13:58:27 is post.craigslist.org/manage/7473121658 ("manage posting") and visit 21 at 13:58:31 is the public listing baltimore.craigslist.org/hab/d/baltimore-cheaper-than-rx-supplements/7473121658.html, titled "cheaper than Rx supplements - health and beauty - by owner", opened from visit 20 (**E06, E07**).

Challenging evidence: the listing category is "health and beauty", the title says "supplements", and the browser history stores no ad text, price, images or location. The records support that an advertisement was created and reached its manage and public pages; they do not show what was offered.

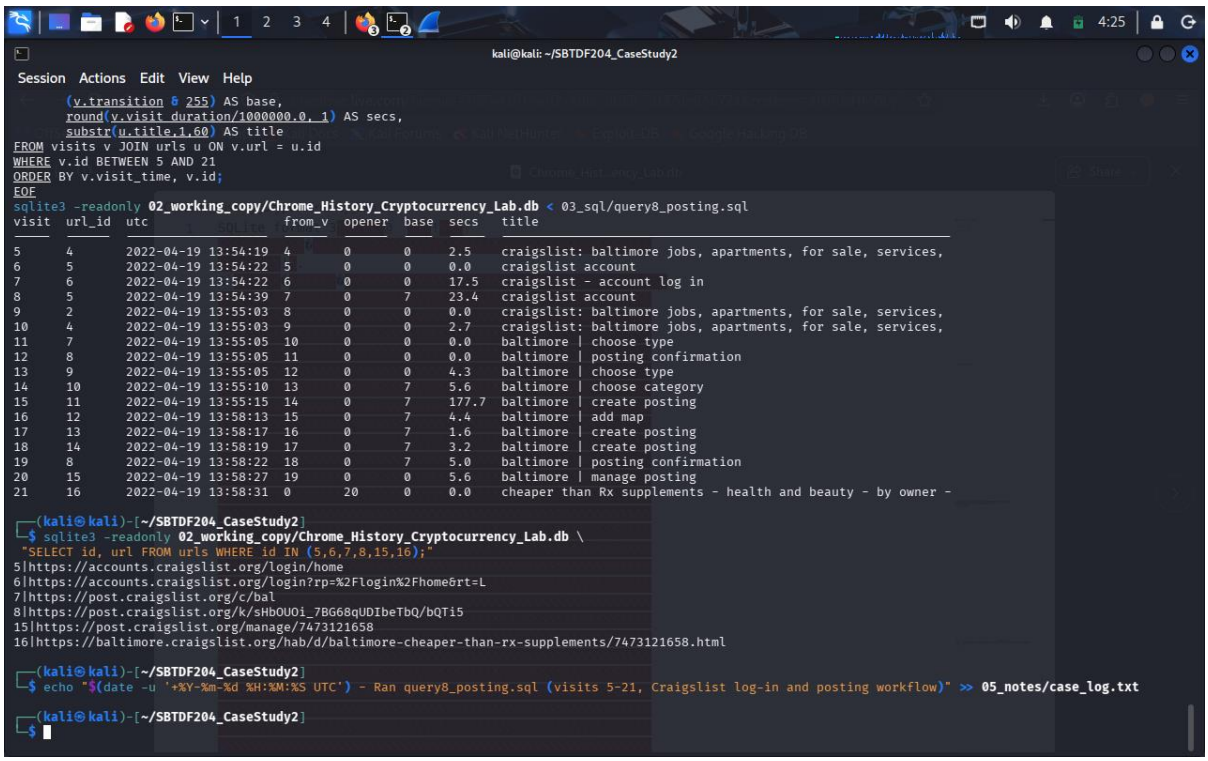


Figure 8 query8_posting.sql*: Craigslist log-in and posting workflow, visits 5–21, with time on page, and the full URLs of the posting pages (E05–E07).*

Q6 Communications and payment activity

- **Gmail access (13:58:38–13:59:31).** Visit 22 searches "gmail"; visits 24–30 form a redirect chain that passes through mail.google.com/mail/ and ends on the public Gmail about page; account chooser, sign-in and password-challenge pages follow (31–36, visit 36 at 13:59:26); visit 37 at 13:59:29 is titled "Inbox (12) - unsub.fscs@gmail.com - Gmail" and visit 42 at 13:59:31 loads #inbox. The account name appears only in page titles (**E08**).
- **Buyer-contact indicator (14:04:12).** Visit 43 loads a Gmail thread titled "cheaper than Rx supplements - order"; visits 44–45 (14:04:35, 14:04:46) add ?compose=… fragments via client redirect and visit 46 returns to the thread at 14:04:52; visit 47 returns to the inbox at 14:05:59. The subject matches the Craigslist title plus "order"; whether the message came from a third party, what it said and whether any reply was sent are not recorded (**E09**).
- **Transaction-ID thread (14:56:47).** Visit 48 loads a thread whose title ends "txid: 517b2156…8afdc5" (the full 64-hex value is in the database; it appears again in visits 55–57, 67–70, 74). Visit 49 (google.com/url?q=https://imgur.com/dTgrkP7&source=gmail…) and visit 50 (imgur.com/dTgrkP7, 18.9 s) show a link in that email opened at 14:56:52 (**E10, E11**).
- **Download id 1 (14:56:58–14:57:10).** C:\Users\FSCS_User\Desktop\proof_of_payment.png, 33,844 of 33,844 bytes, state 1 (complete), referrer https://imgur.com/, tab URL https://imgur.com/dTgrkP7, URL chain https://i.imgur.com/dTgrkP7.png, last_modified header "Tue, 19 Apr 2022 14:52:06 GMT", MIME image/png, opened 0, hash field empty. Chrome does not store the image; the filename does not prove what the image shows, and could have been set by the server or the user (**E12**).
- **Blockchain explorers (14:58:59–14:59:21).** Visit 51 searches "blockchain explorer"; visit 53–54 load blockchain.com/explorer; visit 55 is blockchain.com/search?search=<txid> redirecting (visit 56) to /btc/tx/<txid> titled "Transaction: 517b2156…"; visit 58 at 14:59:21 is /btc/address/38RcsURWCDmCbYocmsZCnGz1FtD7x477mt. The

address page may be an address shown on the transaction page, but the database does not say which role it plays (**E13**).

- **Kraken session (14:59:38–15:02:05).** Visit 59 searches "kraken"; sign-in (61–62, 14:59:47); "Approve new device" (63, 15:00:21, **104.4 s** on page); trade/instant/funding pages (64–66, 15:01:11–15:01:21); the funding title reads "41,543.80 USD - Kraken - Funding - Overview", and the ledger (visit 71, 15:02:05) "41,563.70 USD - Kraken - History - Ledger" (**E14**).
- **Second explorer via Kraken (15:01:31–15:01:33).** Visit 67 is kraken.com/redirect?url=https%3A%2F%2Fmempool.space%2Ftx%2F<txid> opened from visit 66 (funding); it passes a client redirect (68) and server redirect (69) and ends at mempool.space/tx/<txid> (70) the same ID as above (**E15**).
- **Return to e-mail (15:02:32–15:02:56).** Visits 72–74 are ?compose=… client-redirect fragments in the txid thread (from_visit 48, 72, 73), ending back on the thread at 15:02:56 (**E16**).

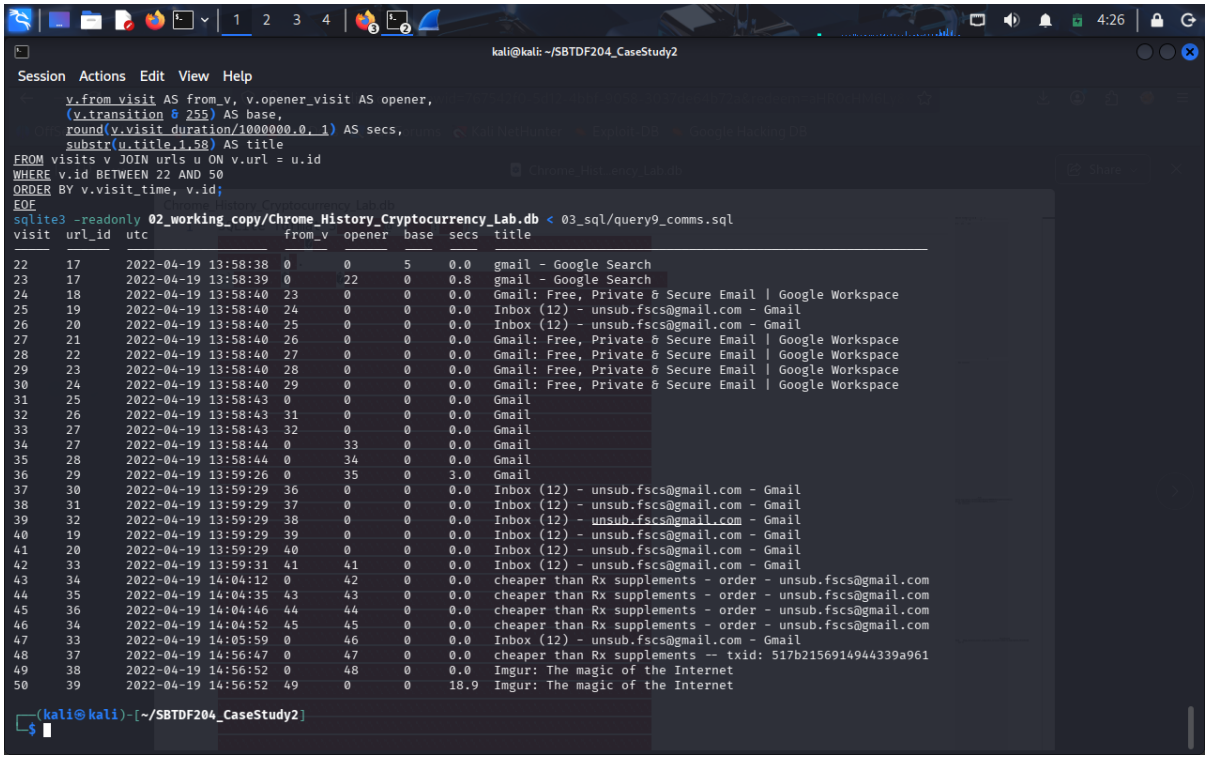


Figure 9 query9_comms.sql*: Gmail access, e-mail threads and the Imgur visit, visits 22–50 (E08–E11).*

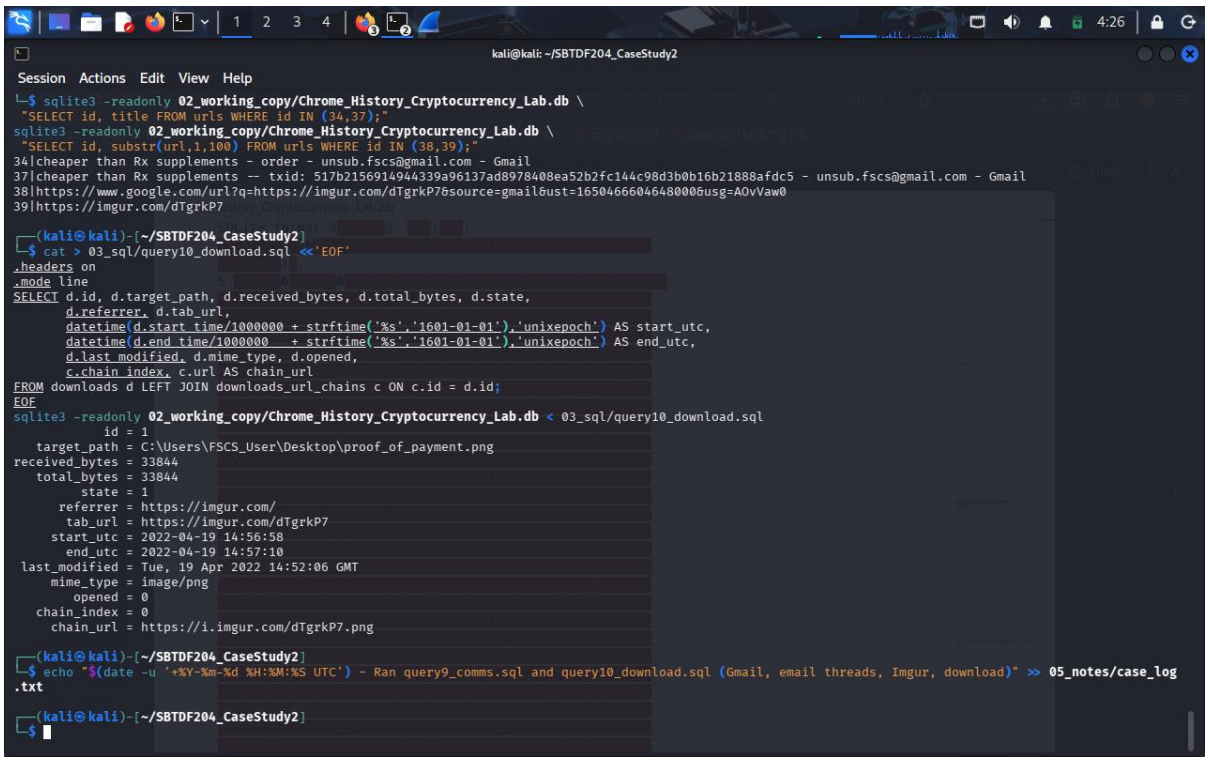


Figure 10 *Full e-mail thread titles (including the transaction ID), the Imgur URLs, and the* query10_download.sql *record for download id 1 (E09–E12).*

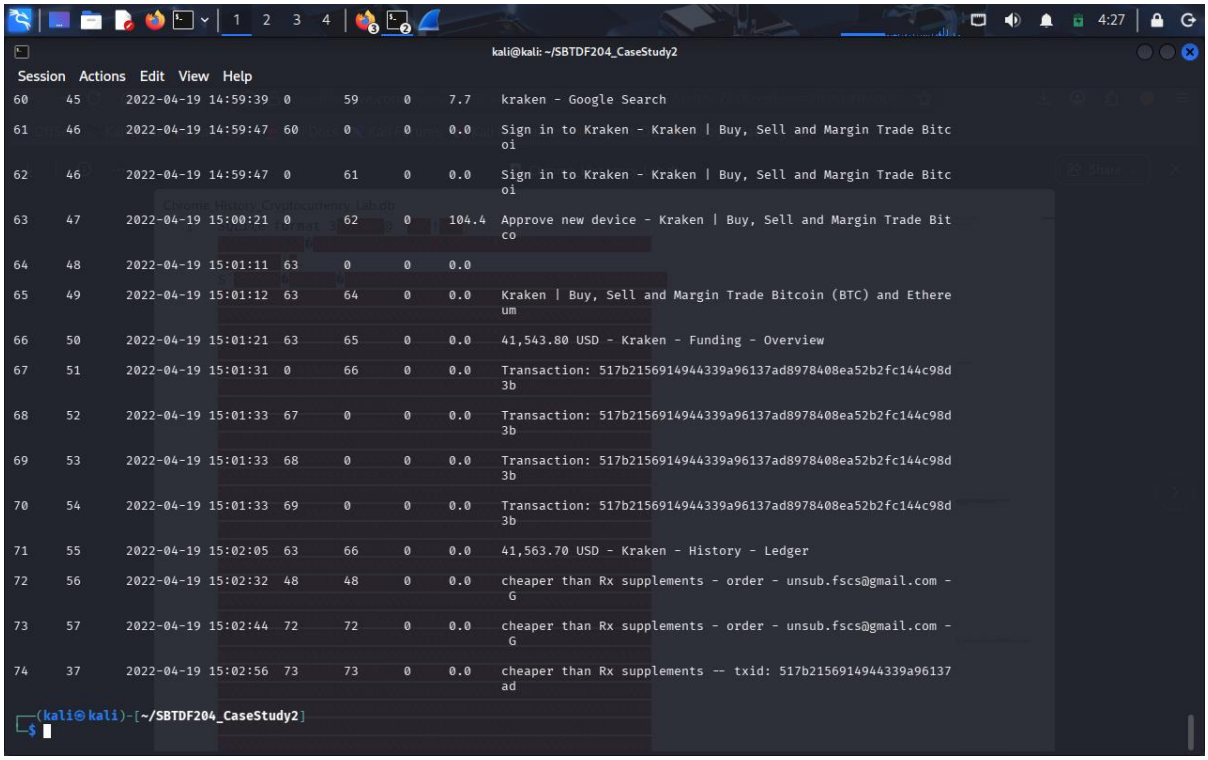


Figure 11 query11_payment.sql*: Kraken pages, the redirect chain to mempool.space and the return to Gmail, visits 60–74 (E14–E16). Visits 51–59 are in the exported* visits_timeline.csv.

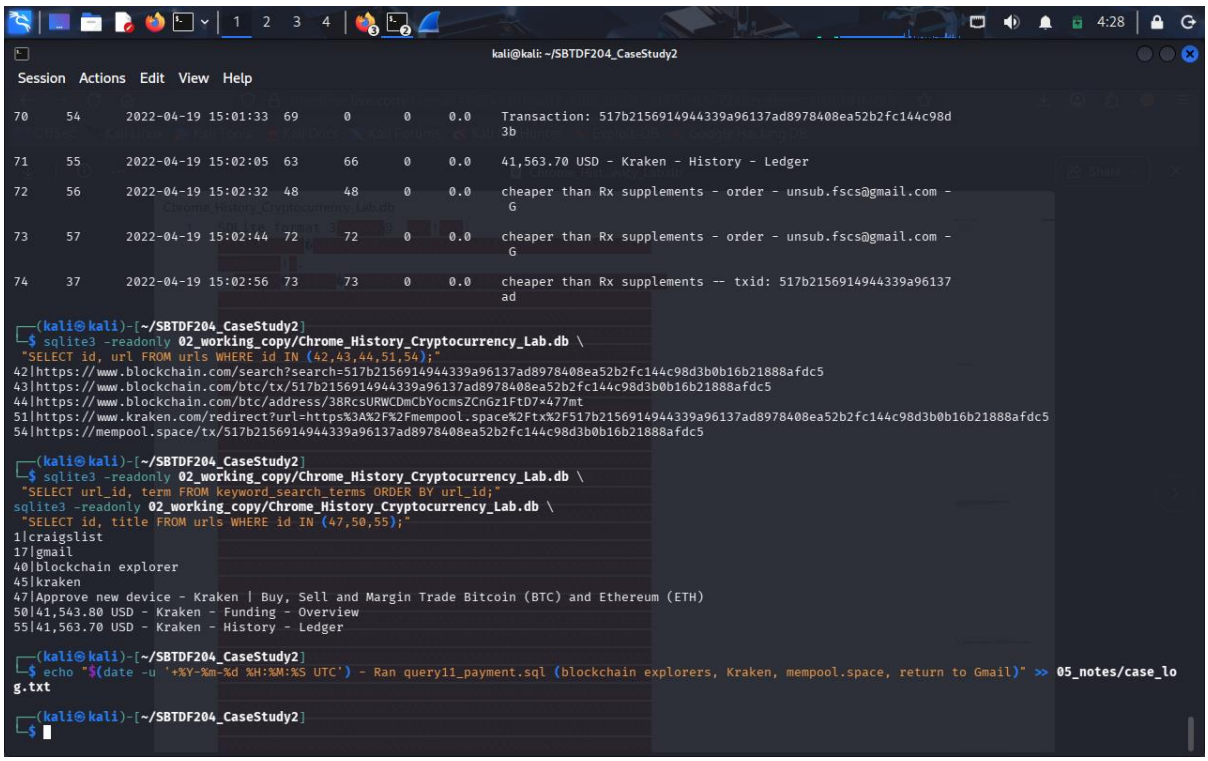


Figure 12 *Full URLs of the explorer and redirect pages (U42, U43, U44, U51, U54), the four search terms, and the Kraken page titles (E04, E13–E15).*

Q7 Reconstructed timeline

See Section 6 (stage-level, observation vs interpretation) and Appendix C (all 74 visits).

Q8 Defensible conclusion

A cautious, evidence-based conclusion is given in Section 2 and the confidence and limitations are in Section 7.

6. Timeline (UTC, 19 April 2022)

"Visit" = visits.id; "URL" = urls.id; transition shown as base type (qualifiers). Observed facts are what the table fields record; interpretation is analytical and labelled with its limit.

Stage	UTC (19 Apr 2022)	Visit / URL ID	Artefact	Observed fact	Interpretation and limit
Research / access	13:54:18	V1 / U1	Google search "craigslist"	Base 5 GENERATED (chain start+end), no from_visit; keyword_search_terms = "craigslist".	Intent to reach Craigslist. Search results content not stored.
Research / access	13:54:19	V2–V5 / U1–U4	Results page again; craigslist.org → geo → baltimore.craigslist.org	V2 link to the same results URL (opener V1); V3 (from V2) chain start; V4 server redirect; V5 server redirect + chain end; V5 on page 2.5 s.	Click-through from the search results, then an automatic regional redirect to the Baltimore site.
Account access	13:54:22	V6–V7 / U5–U6	accounts.craigslist.org/login/home → log-in page	V6 chain start; V7 server redirect + chain end, title "craigslist - account log in", 17.5 s on page.	Browser was not already signed in to Craigslist at this moment.
Account access	13:54:39	V8 / U5	"craigslist account"	Base 7 FORM_SUBMIT, from_visit 7; 23.4 s on page.	Consistent with log-in form submission. Credentials/outcome not recorded; title shows no account name.
Posting	13:55:03–13:55:05	V9–V13 / U2,U4,U7–U9	Return to baltimore home; post.craigslist.org/c/bal → draft key URL → ?s=type	V9 link, V10 server redirect; V11 chain start ("choose type"); V12 server redirect to key URL titled "posting confirmation"; V13 server redirect + chain end ?s=type.	Start of posting wizard. Title at V12 is a generic page name, not proof of a published ad.
Posting	13:55:10	V14 / U10	?s=cat "choose category"	Base 7 FORM_SUBMIT, from_visit 13; 5.6 s.	Posting type submitted; category chosen next.
Posting	13:55:15	V15 / U11	?s=edit "create posting"	Base 7 FORM_SUBMIT, from_visit 14; **177.7 s** on page.	Time consistent with composing the ad text (content not stored).
Posting	13:58:13–13:58:19	V16–V18 / U12–U14	?s=geoverify "add map"; ?s=editimage; ?s=preview	Three consecutive FORM_SUBMITs (from_visit 15, 16, 17); 4.4 s, 1.6 s, 3.2 s.	Map, image and preview steps completed in rapid sequence. Whether images were attached is not recorded.
Posting	13:58:22	V19 / U8	"posting confirmation"	Base 7 FORM_SUBMIT, from_visit 18; 5.0 s.	Posting submitted to Craigslist. Same URL as V12, so the later form-submit transition is what indicates completion.
Posting	13:58:27	V20 / U15	post.craigslist.org/manage/7473121658	Link, from_visit 19; 5.6 s.	Posting management page for ad 7473121658.
Posting	13:58:31	V21 / U16	Listing "cheaper than Rx supplements - health and beauty - by owner"	Link, from_visit 0, opener 20 (new tab/window).	Public listing viewed from the manage page. Listing text/price/images not stored; "Rx" is only a title word.

Stage	UTC (19 Apr 2022)	Visit / URL ID	Artefact	Observed fact	Interpretation and limit
Communications	13:58:38–13:58:40	V22–V30 / U17–U24	Search "gmail"; google.com/gmail → mail.google.com → … → Gmail about page	V22 base 5 + address bar; V23 link (opener V22); V24 chain start; V25–V29 server redirects; V30 server redirect + chain end (www.google.com/gmail/about/).	The attempt to open the inbox ended on the public about page, consistent with no active Gmail session at that moment.
Communications	13:58:43–13:59:31	V31–V42 / U25–U33	Account chooser → sign-in → password challenge → CheckCookie → inbox	V36 signin/v2/challenge/pwd at 13:59:26; V37 title "Inbox (12) - unsub.fscs@gmail.com - Gmail" at 13:59:29; V42 #inbox at 13:59:31.	Consistent with a successful Gmail sign-in for the account named in the titles. Authentication result inferred from titles only.
Communications	14:04:12–14:05:59	V43–V47 / U34–U36, U33	Gmail thread "cheaper than Rx supplements - order"; compose fragments; inbox	V43 chain start (opener V42); V44–V45 client redirects with ?compose=; V46 client redirect + chain end; V47 #inbox.	Subject links the thread to the listing. Sender, text, and whether any reply was sent are not recorded.
Gap	14:05:59 → 14:56:47	—	No visits for ≈ 50.8 min	V47 → V48 time difference 3,048.7 s.	Unobserved period. Offline, other browser/profile, or deleted history all possible.
Payment evidence	14:56:47	V48 / U37	Gmail thread titled "…-- txid: 517b2156…8afdc5"	Chain start, opener V47. Title contains a 64-hex value.	Message carrying a transaction identifier, presented in the context of the same listing subject. Direction and wording unknown.
Payment evidence	14:56:52	V49–V50 / U38–U39	google.com/url?q=imgur.com/dTgrkP7 → imgur.com/dTgrkP7	V49 chain start (opener V48, source=gmail); V50 server redirect + chain end, 18.9 s.	Link in the e-mail was opened. Image content not stored.
Payment evidence	14:56:58–14:57:10	Download id 1	proof_of_payment.png on Desktop (C:\Users\FSCS_User)	33,844/33,844 bytes; state 1; referrer imgur.com; chain i.imgur.com/dTgrkP7.png; last_modified 14:52:06 GMT; opened 0; hash blank.	Image saved locally; its filename suggests it was treated as proof of payment. The file itself was not examined and the name does not prove its content.
Verification	14:58:59–14:59:02	V51–V54 / U40–U41	Search "blockchain explorer"; blockchain.com/explorer	V51 base 5 + address bar; V52 link; V53–V54 explorer page (V54 from_visit 0, opener 53; 18.7 s).	Intent to check a transaction publicly.
Verification	14:59:06	V55–V57 / U42–U43	blockchain.com/search?search=<txid> → /btc/tx/<txid>	V55 chain start (from V54); V56 server redirect + chain end; V57 same tx URL (from V54, opener V56). Titles "Transaction: 517b2156…".	The txid from the e-mail subject was looked up. Amount, confirmations and addresses not stored.
Verification	14:59:21	V58 / U44	blockchain.com/btc/address/38RcsURWCDmCbYocmsZCnGz1FtD7x477mt	Link, from_visit 54, opener 57.	Address page opened after the transaction page; its role (sender/recipient/user wallet) is not shown.
Exchange	14:59:38–15:00:21	V59–V63 / U45–U47	Search "kraken"; sign-in; "Approve new device"	V59 base 5 + address bar; V61–V62 sign-in; V63 device-approval, **104.4 s** on page.	Consistent with signing in to a Kraken account from an unrecognised device. Account owner not shown.

Stage	UTC (19 Apr 2022)	Visit / URL ID	Artefact	Observed fact	Interpretation and limit
Exchange	15:01:11–15:01:21	V64–V66 / U48–U50	kraken.com/u/trade, /u/instant, /u/funding	Links from V63; V66 title "41,543.80 USD - Kraken - Funding - Overview".	Account holds a USD-denominated balance as displayed in the title. Composition and fluctuation unknown.
Verification	15:01:31–15:01:33	V67–V70 / U51–U54	kraken.com/redirect?url=…mempool.space/tx/<txid> → mempool.space/tx/<txid>	V67 chain start (opener V66); V68 client redirect; V69 server redirect; V70 server redirect + chain end; title "Transaction: 517b2156… - mempool - Bitcoin Explorer".	Same txid checked in a second explorer reached from an exchange link.
Exchange	15:02:05	V71 / U55	kraken.com/u/history/ledger	Link, from_visit 63, opener 66; title "41,563.70 USD - Kraken - History - Ledger".	Ledger reviewed. Title balance differs by +19.90 USD vs V66 see Section 7 (not evidence of a credit).
Communications	15:02:32–15:02:56	V72–V74 / U56,U57,U37	Gmail txid thread with ?compose= fragments	V72–V73 client redirects (from_visit 48, 72); V74 client redirect + chain end on the thread at 15:02:56.	Return to the transaction-ID thread after verification, with compose activity. Whether a message was sent is unknown.

7. Attribution assessment

Overall confidence.

Proposition	Confidence	Basis
This Chrome profile executed the Craigslist posting workflow, then Gmail, Imgur, explorer and Kraken activity, in this order	Moderate–high	Internally consistent from_visit/opener_visit chains, consistent timestamps, a form-submit sequence, and a download linked to the Imgur visit.
The activity amounts to advertising and selling a controlled drug	Low–moderate	Only evidence is the title "cheaper than Rx supplements" (health and beauty). Page content, e-mail content and the product are not stored.
A payment was made, received and confirmed	Low–moderate	Txid in an e-mail subject, an image named proof_of_payment, and the same txid in two explorers show checking activity, not outcome.
The activity was carried out by a specific named person	Low	Account name unsub.fscs@gmail.com (titles) and the path C:\Users\FSCS_User\ identify a profile/account, not a human.

Limitations and alternative explanations.

- **Browser profile ≠ human identity.** History shows what a profile loaded. Anyone with access to the logged-in profile, the Windows account FSCS_User or the Gmail/Craigslist/Kraken sessions could produce it; no biometric, camera or login-challenge outcome is recorded.
- **Shared or remote use.** A shared computer, remote access, or a second person using the same profile would look identical in the database. The Kraken "Approve new device" page suggests the exchange did not recognise the device, but not who approved it.
- **Redirects and automatic loads.** Many rows are server/client redirects (22 visits carry a server-redirect bit; 8 carry a client-redirect bit), which are generated by sites and not by user clicks. Titles such as "posting confirmation" appear on the draft URL before final submission.
- **Missing and cached content.** The database holds URLs, titles and timing only. No Craigslist ad text, e-mail body, Imgur image, explorer output (amounts, addresses, confirmations) or exchange balances beyond titles are stored. Page titles can be cached or out of date.
- **Ambiguous balance figures.** Kraken titles show 41,543.80 USD (V66) and 41,563.70 USD (V71). A +19.90 USD change could reflect a credited deposit, but a displayed total portfolio value also moves with market prices, so it is not evidence of a payment.
- **Titles are stored per URL, not per visit.** urls.title holds the most recent title for a URL, so an earlier visit can display a later title (for example, visits 25–26 and 30–32 show "Inbox (12) - …" although they are redirect hops before the 13:59:29 sign-in). Titles were therefore treated as supporting, not decisive, evidence.
- **Gap and completeness.** A ≈50.8-minute gap (14:05:59–14:56:47) is unobserved; other browsers, profiles, devices, private/incognito windows or deleted history entries would not appear. visit_source is empty, which Chrome treats as locally browsed, so synced-device activity is not indicated.
- **Timestamp context.** Times are device-reported and converted to UTC; no clock verification was possible. The local-time offset of the device is unknown.
- **Hypothesis bias.** The case brief frames the activity as drug trafficking. The records also fit an ordinary online sale of legal supplements paid for in Bitcoin; the history cannot distinguish the two, so corroboration (device image, page caches, e-mail content, the downloaded image, blockchain data obtained lawfully) is needed.

8. Appendices

Appendix A SQL used

Each query was saved as a .sql file in 03_sql/ and run with sqlite3 -readonly 02_working_copy/Chrome_History_Cryptocurrency_Lab.db < 03_sql/<file>. The .headers on and .mode lines set the output format. Inline queries are shown last.

query1_schema.sql tables, schemas, integrity

```
SELECT name FROM sqlite_master WHERE type='table' ORDER BY name;
PRAGMA table_info(urls);  PRAGMA table_info(visits);
PRAGMA table_info(downloads);  PRAGMA table_info(downloads_url_chains);
PRAGMA integrity_check;
```

query2_counts.sql row counts

```
SELECT 'urls' AS tbl, COUNT(*) AS rows FROM urls
UNION ALL SELECT 'visits', COUNT(*) FROM visits
UNION ALL SELECT 'downloads', COUNT(*) FROM downloads
UNION ALL SELECT 'downloads_url_chains', COUNT(*) FROM downloads_url_chains
UNION ALL SELECT 'keyword_search_terms', COUNT(*) FROM keyword_search_terms;
```

query3_timerange.sql first and last visit (UTC)

```
SELECT datetime(MIN(visit_time)/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS first_utc,
       datetime(MAX(visit_time)/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS last_utc,
       COUNT(*) AS total_visits FROM visits;
```

query4_timeline.sql full timeline exported to CSV

```
.mode csv
.output 05_notes/visits_timeline.csv
SELECT v.id AS visit_id, u.id AS url_id,
  datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS visit_utc,
  v.from_visit, v.opener_visit, v.transition, (v.transition & 255) AS base_type,
  v.visit_duration/1000.0 AS duration_ms, u.title, u.url
FROM visits v JOIN urls u ON v.url = u.id ORDER BY v.visit_time, v.id;
.output stdout
```

query5_transitions.sql distinct transition values

```
SELECT transition, printf('0x%08X', transition & 4294967295) AS hex,
  (transition & 255) AS base_type, COUNT(*) AS n, GROUP_CONCAT(id) AS visit_ids
FROM visits GROUP BY transition ORDER BY MIN(id);
```

query6_chains.sql visit-to-parent chain

```
SELECT v.id AS visit, v.from_visit AS from_v, v.opener_visit AS opener,
  printf('0x%08X', v.transition & 4294967295) AS hex, (v.transition & 255) AS base,
  substr(u.url,1,50) AS url, substr(pu.url,1,35) AS came_from
FROM visits v JOIN urls u ON v.url = u.id
LEFT JOIN visits pv ON v.from_visit = pv.id LEFT JOIN urls pu ON pv.url = pu.id
ORDER BY v.visit_time, v.id;
```

query7_decode.sql qualifier flags

```
SELECT printf('0x%08X', transition & 4294967295) AS hex, (transition & 255) AS base,
  CASE transition & 255 WHEN 0 THEN 'LINK' WHEN 1 THEN 'TYPED' WHEN 5 THEN 'GENERATED'
       WHEN 7 THEN 'FORM_SUBMIT' ELSE 'OTHER' END AS base_name,
  (transition & 33554432) > 0 AS addr_bar, (transition & 268435456) > 0 AS chain_start,
  (transition & 536870912) > 0 AS chain_end, (transition & 1073741824) > 0 AS client_redir,
  (transition & 2147483648) > 0 AS server_redir, COUNT(*) AS n
FROM visits GROUP BY transition ORDER BY MIN(id);
```

query8_posting.sql Craigslist log-in and posting (visits 5–21)

```
SELECT v.id AS visit, u.id AS url_id,
  datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS utc,
  v.from_visit AS from_v, v.opener_visit AS opener, (v.transition & 255) AS base,
  round(v.visit_duration/1000000.0, 1) AS secs, substr(u.title,1,60) AS title
FROM visits v JOIN urls u ON v.url = u.id
WHERE v.id BETWEEN 5 AND 21 ORDER BY v.visit_time, v.id;
```

query9_comms.sql Gmail, e-mail threads, Imgur (visits 22–50)

```
-- same SELECT as query8_posting.sql, with substr(u.title,1,58) and:
WHERE v.id BETWEEN 22 AND 50 ORDER BY v.visit_time, v.id;
```

query10_download.sql download and URL chain

```
.mode line
SELECT d.id, d.target_path, d.received_bytes, d.total_bytes, d.state, d.referrer, d.tab_url,
  datetime(d.start_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS start_utc,
  datetime(d.end_time/1000000   + strftime('%s','1601-01-01'),'unixepoch') AS end_utc,
  d.last_modified, d.mime_type, d.opened, c.chain_index, c.url AS chain_url
FROM downloads d LEFT JOIN downloads_url_chains c ON c.id = d.id;
```

query11_payment.sql explorers, Kraken, return to Gmail (visits 51–74)

```
-- same SELECT as query9_comms.sql, with substr(u.title,1,62) and:
WHERE v.id BETWEEN 51 AND 74 ORDER BY v.visit_time, v.id;
```

Inline queries full titles, URLs and search terms

```
SELECT id, title FROM urls WHERE id IN (34,37);
SELECT id, substr(url,1,100) FROM urls WHERE id IN (38,39);
SELECT id, url FROM urls WHERE id IN (5,6,7,8,15,16);
SELECT id, url FROM urls WHERE id IN (42,43,44,51,54);
SELECT url_id, term FROM keyword_search_terms ORDER BY url_id;
SELECT id, title FROM urls WHERE id IN (47,50,55);
```

Appendix B Screenshot index

All screenshots were taken from the examiner's Kali terminal while the queries in Appendix A were run. They are placed beside the finding they support in Sections 3 and 5.

Figure	What it shows	Evidence ID	Placed in
1	Assignment folder structure, first case-log entry, tools installed	E01	Section 3
1a	Source file in ~/Downloads and the failed first copy attempt (placeholder file name)	E01	Section 3
2	Original copied read-only; size; SHA-256 of original and working copy	E01	Section 3
3	End of schema listing, integrity check, row counts	E02	Q2
3a	Earlier part of the schema listing (visits and downloads fields)	E02	Q2
4	Timeline export to CSV; first visit 13:54:18 UTC	E03	Q3
5	Distinct transition values	E04	Q4
6	from_visit / opener chains, visits 1–38	E04–E06	Q4
7	Qualifier flags per transition value	E04	Q4
8	Craigslist log-in and posting workflow (visits 5–21); posting URLs	E05–E07	Q5
9	Gmail, e-mail threads, Imgur (visits 22–50)	E08–E11	Q6
10	Full thread titles, Imgur URLs, download record	E09–E12	Q6
11	Kraken, mempool.space chain, return to Gmail (visits 60–74)	E14–E16	Q6
12	Explorer/redirect URLs, search terms, Kraken titles	E04, E13–E15	Q6

Appendix C Visit register (all 74 visits, UTC date 2022-04-19)

Columns: V = visits.id; U = urls.id; "From" = from_visit; "Open" = opener_visit; Stored = visits.transition as returned by SQLite; Base/qualifiers decoded as in Section 4.3; Dur = visit_duration in seconds. URLs shortened; 64-hex transaction IDs are shown as the first 8 characters + "…".

V	U	UTC	From	Open	Stored	Base + qualifiers	Dur (s)	URL (short)	Title
1	1	13:54:18	0	0	805306373	5 GENERATED + CHAIN_START + CHAIN_END	0	www.google.com/search?q=craigslist&rlz=1C1ONGR_enUS989US989&…	craigslist - Google Search
2	1	13:54:19	0	1	805306368	0 LINK + CHAIN_START + CHAIN_END	0.7	www.google.com/search?q=craigslist&rlz=1C1ONGR_enUS989US989&…	craigslist - Google Search
3	2	13:54:19	2	0	268435456	0 LINK + CHAIN_START	0	www.craigslist.org/	craigslist: baltimore jobs, apartments, for sale, services, community,
4	3	13:54:19	3	0	-2147483648	0 LINK + SERVER_REDIRECT	0	geo.craigslist.org/	craigslist: baltimore jobs, apartments, for sale, services, community,
5	4	13:54:19	4	0	-1610612736	0 LINK + CHAIN_END + SERVER_REDIRECT	2.5	baltimore.craigslist.org/	craigslist: baltimore jobs, apartments, for sale, services, community,
6	5	13:54:22	5	0	268435456	0 LINK + CHAIN_START	0	accounts.craigslist.org/login/home	craigslist account
7	6	13:54:22	6	0	-1610612736	0 LINK + CHAIN_END + SERVER_REDIRECT	17.5	accounts.craigslist.org/login?rp=%2Flogin%2Fhome&rt=L	craigslist - account log in
8	5	13:54:39	7	0	805306375	7 FORM_SUBMIT + CHAIN_START + CHAIN_END	23.4	accounts.craigslist.org/login/home	craigslist account
9	2	13:55:03	8	0	268435456	0 LINK + CHAIN_START	0	www.craigslist.org/	craigslist: baltimore jobs, apartments, for sale, services, community,
10	4	13:55:03	9	0	-1610612736	0 LINK + CHAIN_END + SERVER_REDIRECT	2.7	baltimore.craigslist.org/	craigslist: baltimore jobs, apartments, for sale, services, community,
11	7	13:55:05	10	0	268435456	0 LINK + CHAIN_START	0	post.craigslist.org/c/bal	baltimore | choose type
12	8	13:55:05	11	0	-2147483648	0 LINK + SERVER_REDIRECT	0	post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5	baltimore | posting confirmation
13	9	13:55:05	12	0	-1610612736	0 LINK + CHAIN_END + SERVER_REDIRECT	4.3	post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=type	baltimore | choose type
14	10	13:55:10	13	0	805306375	7 FORM_SUBMIT + CHAIN_START + CHAIN_END	5.6	post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=cat	baltimore | choose category
15	11	13:55:15	14	0	805306375	7 FORM_SUBMIT + CHAIN_START + CHAIN_END	177.7	post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=edit	baltimore | create posting
16	12	13:58:13	15	0	805306375	7 FORM_SUBMIT + CHAIN_START + CHAIN_END	4.4	post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=geoverify	baltimore | add map
17	13	13:58:17	16	0	805306375	7 FORM_SUBMIT + CHAIN_START + CHAIN_END	1.6	post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=editimage	baltimore | create posting
18	14	13:58:19	17	0	805306375	7 FORM_SUBMIT + CHAIN_START + CHAIN_END	3.2	post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=preview	baltimore | create posting

V	U	UTC	From	Open	Stored	Base + qualifiers	Dur (s)	URL (short)	Title
19	8	13:58:22	18	0	805306375	7 FORM_SUBMIT + CHAIN_START + CHAIN_END	5	post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5	baltimore | posting confirmation
20	15	13:58:27	19	0	805306368	0 LINK + CHAIN_START + CHAIN_END	5.6	post.craigslist.org/manage/7473121658	baltimore | manage posting
21	16	13:58:31	0	20	805306368	0 LINK + CHAIN_START + CHAIN_END	0	baltimore.craigslist.org/hab/d/baltimore-cheaper-than-rx-sup…	cheaper than Rx supplements - health and beauty - by owner -...
22	17	13:58:38	0	0	838860805	5 GENERATED + FROM_ADDRESS_BAR + CHAIN_START + CHAIN_END	0	www.google.com/search?q=gmail&rlz=1C1ONGR_enUS989US989&oq=gm…	gmail - Google Search
23	17	13:58:39	0	22	805306368	0 LINK + CHAIN_START + CHAIN_END	0.8	www.google.com/search?q=gmail&rlz=1C1ONGR_enUS989US989&oq=gm…	gmail - Google Search
24	18	13:58:40	23	0	268435456	0 LINK + CHAIN_START	0	www.google.com/gmail/	Gmail: Free, Private & Secure Email | Google Workspace
25	19	13:58:40	24	0	-2147483648	0 LINK + SERVER_REDIRECT	0	mail.google.com/mail/	Inbox (12) - unsub.fscs@gmail.com - Gmail
26	20	13:58:40	25	0	-2147483648	0 LINK + SERVER_REDIRECT	0	mail.google.com/mail/u/0/	Inbox (12) - unsub.fscs@gmail.com - Gmail
27	21	13:58:40	26	0	-2147483648	0 LINK + SERVER_REDIRECT	0	accounts.google.com/ServiceLogin?service=mail&passive=120960…	Gmail: Free, Private & Secure Email | Google Workspace
28	22	13:58:40	27	0	-2147483648	0 LINK + SERVER_REDIRECT	0	mail.google.com/intl/en/mail/help/about.html	Gmail: Free, Private & Secure Email | Google Workspace
29	23	13:58:40	28	0	-2147483648	0 LINK + SERVER_REDIRECT	0	www.google.com/intl/en/mail/help/about.html	Gmail: Free, Private & Secure Email | Google Workspace
30	24	13:58:40	29	0	-1610612736	0 LINK + CHAIN_END + SERVER_REDIRECT	0	www.google.com/gmail/about/	Gmail: Free, Private & Secure Email | Google Workspace
31	25	13:58:43	0	0	268435456	0 LINK + CHAIN_START	0	accounts.google.com/AccountChooser/signinchooser?service=mai…	Gmail
32	26	13:58:43	31	0	-2147483648	0 LINK + SERVER_REDIRECT	0	accounts.google.com/AccountChooser?service=mail&continue=htt…	Gmail
33	27	13:58:43	32	0	-1610612736	0 LINK + CHAIN_END + SERVER_REDIRECT	0	accounts.google.com/ServiceLogin?continue=https%3A%2F%2Fmail…	Gmail
34	27	13:58:44	0	33	805306368	0 LINK + CHAIN_START + CHAIN_END	0	accounts.google.com/ServiceLogin?continue=https%3A%2F%2Fmail…	Gmail
35	28	13:58:44	0	34	805306368	0 LINK + CHAIN_START + CHAIN_END	0	accounts.google.com/signin/v2/identifier?continue=https%3A%2…	Gmail
36	29	13:59:26	0	35	805306368	0 LINK + CHAIN_START + CHAIN_END	3	accounts.google.com/signin/v2/challenge/pwd?continue=https%3…	Gmail
37	30	13:59:29	36	0	268435456	0 LINK + CHAIN_START	0	accounts.google.com/CheckCookie?continue=https%3A%2F%2Fmail.…	Inbox (12) - unsub.fscs@gmail.com - Gmail
38	31	13:59:29	37	0	-2147483648	0 LINK + SERVER_REDIRECT	0	mail.google.com/accounts/SetOSID?authuser=0&continue=https%3…	Inbox (12) - unsub.fscs@gmail.com - Gmail
39	32	13:59:29	38	0	-2147483648	0 LINK + SERVER_REDIRECT	0	accounts.youtube.com/accounts/SetSID?ssdc=1&sidt=ALWU2ctdRyP…	Inbox (12) - unsub.fscs@gmail.com - Gmail
40	19	13:59:29	39	0	-2147483648	0 LINK + SERVER_REDIRECT	0	mail.google.com/mail/	Inbox (12) - unsub.fscs@gmail.com - Gmail
41	20	13:59:29	40	0	-2147483648	0 LINK + SERVER_REDIRECT	0	mail.google.com/mail/u/0/	Inbox (12) - unsub.fscs@gmail.com - Gmail
42	33	13:59:31	41	41	1610612736	0 LINK + CHAIN_END + CLIENT_REDIRECT	0	mail.google.com/mail/u/0/#inbox	Inbox (12) - unsub.fscs@gmail.com - Gmail
43	34	14:04:12	0	42	268435456	0 LINK + CHAIN_START	0	mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqXBGCnHkHGxZnZbk…	cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail

V	U	UTC	From	Open	Stored	Base + qualifiers	Dur (s)	URL (short)	Title
44	35	14:04:35	43	43	1073741824	0 LINK + CLIENT_REDIRECT	0	mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqXBGCnHkHGxZnZbk…	cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail
45	36	14:04:46	44	44	1073741824	0 LINK + CLIENT_REDIRECT	0	mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqXBGCnHkHGxZnZbk…	cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail
46	34	14:04:52	45	45	1610612736	0 LINK + CHAIN_END + CLIENT_REDIRECT	0	mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqXBGCnHkHGxZnZbk…	cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail
47	33	14:05:59	0	46	805306368	0 LINK + CHAIN_START + CHAIN_END	0	mail.google.com/mail/u/0/#inbox	Inbox (12) - unsub.fscs@gmail.com - Gmail
48	37	14:56:47	0	47	268435456	0 LINK + CHAIN_START	0	mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqvtRjhGdkMBqVNFn…	cheaper than Rx supplements -- txid: 517b2156914944339a96137ad8978408e
49	38	14:56:52	0	48	268435456	0 LINK + CHAIN_START	0	www.google.com/url?q=https://imgur.com/dTgrkP7&source=gmail&…	Imgur: The magic of the Internet
50	39	14:56:52	49	0	-1610612736	0 LINK + CHAIN_END + SERVER_REDIRECT	18.9	imgur.com/dTgrkP7	Imgur: The magic of the Internet
51	40	14:58:59	0	0	838860805	5 GENERATED + FROM_ADDRESS_BAR + CHAIN_START + CHAIN_END	0	www.google.com/search?q=blockchain+explorer&rlz=1C1ONGR_enUS…	blockchain explorer - Google Search
52	40	14:59:00	0	51	805306368	0 LINK + CHAIN_START + CHAIN_END	1	www.google.com/search?q=blockchain+explorer&rlz=1C1ONGR_enUS…	blockchain explorer - Google Search
53	41	14:59:01	52	0	805306368	0 LINK + CHAIN_START + CHAIN_END	0	www.blockchain.com/explorer	Blockchain Explorer - Search the Blockchain | BTC | ETH | BCH
54	41	14:59:02	0	53	805306368	0 LINK + CHAIN_START + CHAIN_END	18.7	www.blockchain.com/explorer	Blockchain Explorer - Search the Blockchain | BTC | ETH | BCH
55	42	14:59:06	54	0	268435456	0 LINK + CHAIN_START	0	www.blockchain.com/search?search=517b2156…	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b218
56	43	14:59:06	55	0	-1610612736	0 LINK + CHAIN_END + SERVER_REDIRECT	0	www.blockchain.com/btc/tx/517b2156…	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b218
57	43	14:59:06	54	56	805306368	0 LINK + CHAIN_START + CHAIN_END	0	www.blockchain.com/btc/tx/517b2156…	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b218
58	44	14:59:21	54	57	805306368	0 LINK + CHAIN_START + CHAIN_END	0	www.blockchain.com/btc/address/38RcsURWCDmCbYocmsZCnGz1FtD7x…	Address: 38RcsURWCDmCbYocmsZCnGz1FtD7x477mt | Blockchain Explorer
59	45	14:59:38	0	0	838860805	5 GENERATED + FROM_ADDRESS_BAR + CHAIN_START + CHAIN_END	0	www.google.com/search?q=kraken&rlz=1C1ONGR_enUS989US989&oq=k…	kraken - Google Search
60	45	14:59:39	0	59	805306368	0 LINK + CHAIN_START + CHAIN_END	7.7	www.google.com/search?q=kraken&rlz=1C1ONGR_enUS989US989&oq=k…	kraken - Google Search
61	46	14:59:47	60	0	805306368	0 LINK + CHAIN_START + CHAIN_END	0	www.kraken.com/en-us/sign-in	Sign in to Kraken - Kraken | Buy, Sell and Margin Trade Bitcoin (BTC)
62	46	14:59:47	0	61	805306368	0 LINK + CHAIN_START + CHAIN_END	0	www.kraken.com/en-us/sign-in	Sign in to Kraken - Kraken | Buy, Sell and Margin Trade Bitcoin (BTC)
63	47	15:00:21	0	62	805306368	0 LINK + CHAIN_START + CHAIN_END	104.4	www.kraken.com/en-us/device-approval	Approve new device - Kraken | Buy, Sell and Margin Trade Bitcoin (BTC)
64	48	15:01:11	63	0	805306368	0 LINK + CHAIN_START + CHAIN_END	0	www.kraken.com/u/trade?i=true	

V	U	UTC	From	Open	Stored	Base + qualifiers	Dur (s)	URL (short)	Title
65	49	15:01:12	63	64	805306368	0 LINK + CHAIN_START + CHAIN_END	0	www.kraken.com/u/instant	Kraken | Buy, Sell and Margin Trade Bitcoin (BTC) and Ethereum (ETH)
66	50	15:01:21	63	65	805306368	0 LINK + CHAIN_START + CHAIN_END	0	www.kraken.com/u/funding	41,543.80 USD - Kraken - Funding - Overview
67	51	15:01:31	0	66	268435456	0 LINK + CHAIN_START	0	www.kraken.com/redirect?url=https%3A%2F%2Fmempool.space%2Ftx…	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b218
68	52	15:01:33	67	0	1073741824	0 LINK + CLIENT_REDIRECT	0	www.kraken.com/redirect?t=-_F22IhwssFD	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b218
69	53	15:01:33	68	0	-2147483648	0 LINK + SERVER_REDIRECT	0	www.kraken.com/redirect?t=-_F22IhwssFD	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b218
70	54	15:01:33	69	0	-1610612736	0 LINK + CHAIN_END + SERVER_REDIRECT	0	mempool.space/tx/517b2156…	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b218
71	55	15:02:05	63	66	805306368	0 LINK + CHAIN_START + CHAIN_END	0	www.kraken.com/u/history/ledger	41,563.70 USD - Kraken - History - Ledger
72	56	15:02:32	48	48	1073741824	0 LINK + CLIENT_REDIRECT	0	mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqvtRjhGdkMBqVNFn…	cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail
73	57	15:02:44	72	72	1073741824	0 LINK + CLIENT_REDIRECT	0	mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqvtRjhGdkMBqVNFn…	cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail
74	37	15:02:56	73	73	1610612736	0 LINK + CHAIN_END + CLIENT_REDIRECT	0	mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqvtRjhGdkMBqVNFn…	cheaper than Rx supplements -- txid: 517b2156914944339a96137ad8978408e

Appendix D Evidence log

Evidence ID	Source record	Finding	Why it matters	SQL / screenshot reference
E01	File Chrome_History_Cryptocurrency_Lab.db (196,608 bytes)	SHA-256 b991b0fa…cde3fd19 for the original and the working copy (identical); original set read-only (mode 444); queries run with sqlite3 -readonly.	Establishes integrity and provenance of the evidence.	Figures 1, 1a, 2; Section 3
E02	sqlite_master; PRAGMA table_info; integrity_check; row counts	7 populated tables; urls 57, visits 74, downloads 1, downloads_url_chains 1, keyword_search_terms 4; integrity ok; history version 53.	Confirms the schema and that the file is not corrupt.	query1, query2; Figures 3, 3a
E03	visits.visit_time (first/last)	First 2022-04-19 13:54:18 UTC (V1); last 15:02:56 UTC (V74); 74 visits.	Defines the investigation window and shows the UTC conversion.	query3, query4; Figures 4, 11
E04	visits V1, V22, V51, V59; transition values; keyword_search_terms	Nine distinct transition values; four base-5 omnibox searches (craigslist, gmail, blockchain explorer, kraken); no base-1 (typed) visits; typed_count = 0.	Shows user-initiated entry points and how each page was reached.	query5, query6, query7; Figures 5–7, 12
E05	visits V6–V8; urls U5–U6	Redirect to log-in page (V7), then FORM_SUBMIT (V8) to accounts.craigslist.org/login/home at 13:54:39.	Consistent with Craigslist account log-in before posting.	query6, query8; Figures 6, 8
E06	visits V11–V19; urls U7–U14, U8	Posting wizard: type → category → edit (177.7 s) → map → image → preview → confirmation; V14–V19 are FORM_SUBMIT in an unbroken from_visit chain, 13:55:05–13:58:22.	Shows a complete multi-step posting workflow.	query8; Figure 8
E07	visits V20–V21; urls U15–U16	Manage page for ad 7473121658 (V20); public listing titled "cheaper than Rx supplements - health and beauty - by owner" opened from V20 (V21).	Identifies the advertisement and its only descriptive text.	query8; Figure 8
E08	visits V22–V42; urls U17–U33	Gmail searched; redirected to the about page; account chooser, sign-in and password challenge; inbox titled with unsub.fscs@gmail.com at 13:59:29.	Shows access to a Gmail account after the listing was created.	query9; Figure 9
E09	visits V43–V47; urls U34–U36	Thread "cheaper than Rx supplements - order" at 14:04:12; compose fragments 14:04:35–14:04:46; inbox at 14:05:59.	Subject ties the thread to the listing; indicates reading or replying to an order-related message.	query9, inline titles; Figures 9, 10
E10	visit V48; url U37	Thread title containing "txid: 517b2156…8afdc5" at 14:56:47, after a ≈ 50.8 min gap.	Introduces the transaction identifier used in all later verification.	query9, inline titles; Figures 9, 10
E11	visits V49–V50; urls U38–U39	Gmail link wrapper (source=gmail) → imgur.com/dTgrkP7 at 14:56:52, 18.9 s on page.	Links the e-mail to the image page that preceded the download.	query9, inline URLs; Figures 9, 10
E12	downloads id 1; downloads_url_chains id 1	C:\Users\FSCS_User\Desktop\proof_of_payment.png; 33,844 B; complete; 14:56:58–14:57:10; chain https://i.imgur.com/dTgrkP7.png; last-modified 14:52:06 GMT.	Shows an image saved locally from the Imgur page, named as proof of payment.	query10; Figure 10
E13	visits V51–V58; urls U40–U44	Explorer search; txid search → /btc/tx/<txid>; address page 38RcsURW…x477mt at 14:59:21.	Shows lookup of the e-mailed transaction ID in a public explorer.	query11, inline URLs; Figure 12, CSV
E14	visits V59–V66, V71; urls U45–U50, U55	Kraken search; sign-in 14:59:47; device approval 15:00:21 (104.4 s); funding title 41,543.80 USD (15:01:21); ledger title 41,563.70 USD (15:02:05).	Shows exchange access and ledger review after the explorer check.	query11, inline titles; Figures 11, 12

Evidence ID	Source record	Finding	Why it matters	SQL / screenshot reference
E15	visits V67–V70; urls U51–U54	kraken.com/redirect?url=…mempool.space/tx/<txid> (opener V66) → client + server redirects → mempool.space/tx/<txid> at 15:01:33.	Shows the same txid checked in a second explorer from the exchange.	query11, inline URLs; Figures 11, 12
E16	visits V72–V74; urls U56–U57, U37	Compose fragments in the txid thread (15:02:32–15:02:56), from_visit 48 → 72 → 73.	Indicates return to the communication thread after verification; sending not proven.	query11; Figure 11
E17	downloads.target_path; urls titles; rlz parameter	Windows user folder FSCS_User; account in titles unsub.fscs@gmail.com; rlz=1C1ONGR_enUS989US989 in Google URLs.	Basis for the attribution assessment (profile/account, not person).	query4, query10; Section 7